

CO3- AT3- Tree Traversal Challenges

Write the traversal sequences for following trees

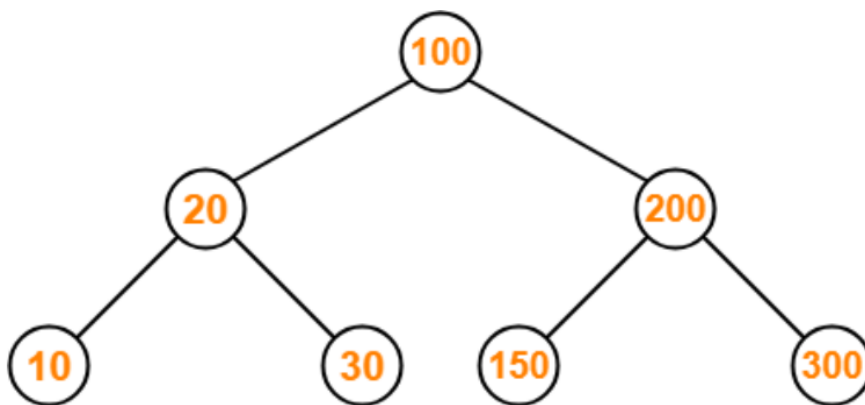
CSA0315-Data Structures

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Date:11-08-2026

Question No :1

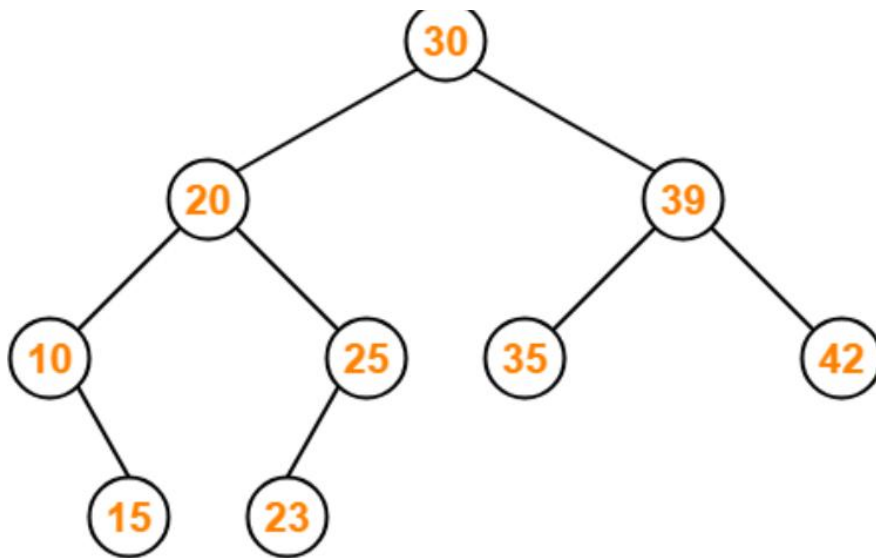


Inorder: 10,20,30,100,150,200,300

Preorder: 100,20,10,30,200,150,300

Postorder: 10,30,20,150,300,200,100

Question No : 2



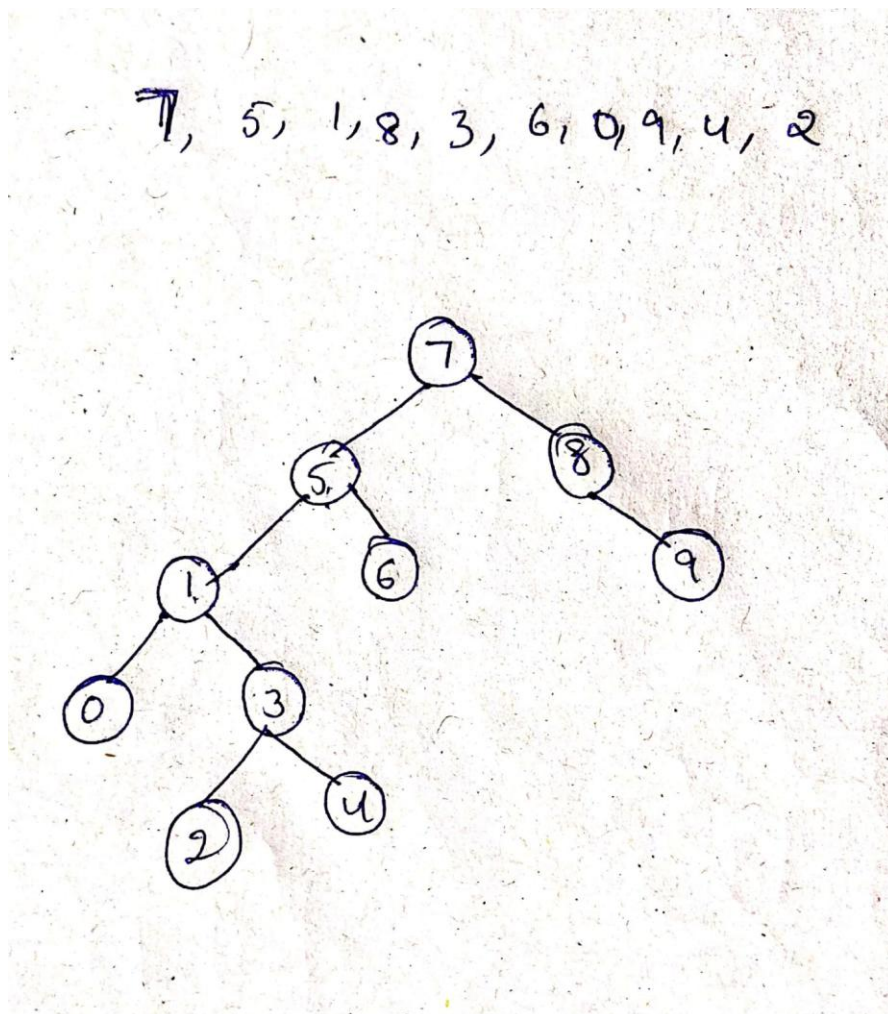
Inorder: 10, 15, 20, 23, 25, 30, 35, 39, 42

Preorder: 30, 20, 10, 15, 25, 23, 39, 35, 42

Postorder: 15, 10, 23, 25, 20, 35, 42, 39, 30

Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



Inorder:0,1,2,3,4,5,6,7,8,9

Preorder:7,5,1,0,3,2,4,6,8,9

Postorder:0,2,4,3,1,6,5,9,8,7

Question No : 5

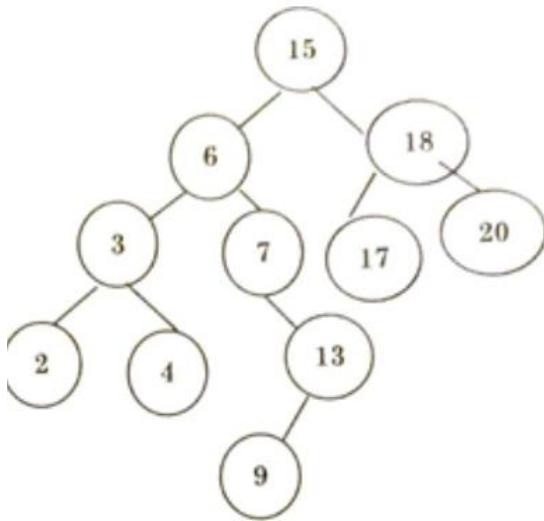


Inorder: 4, 2, 7, 5, 1, 3, 6

Preorder: 1, 2, 4, 5, 7, 3, 6

Postorder: 4, 7, 5, 2, 6, 3, 1

Question No 6



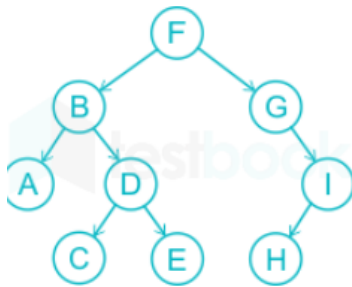
Inorder: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

Preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Postorder: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Ans) Option 3: A,B,C,D,E,F,G,H,I

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

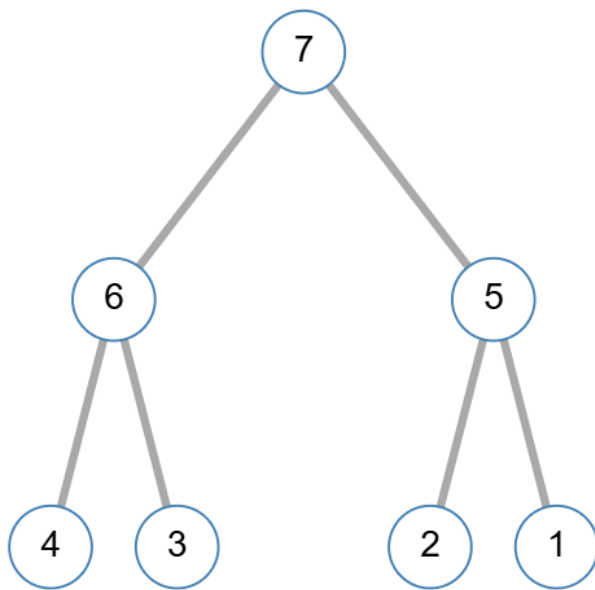
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

Ans) Option 3: D F E B H G C A

Question No 9 : Find Inorder, Preorder and Postorder

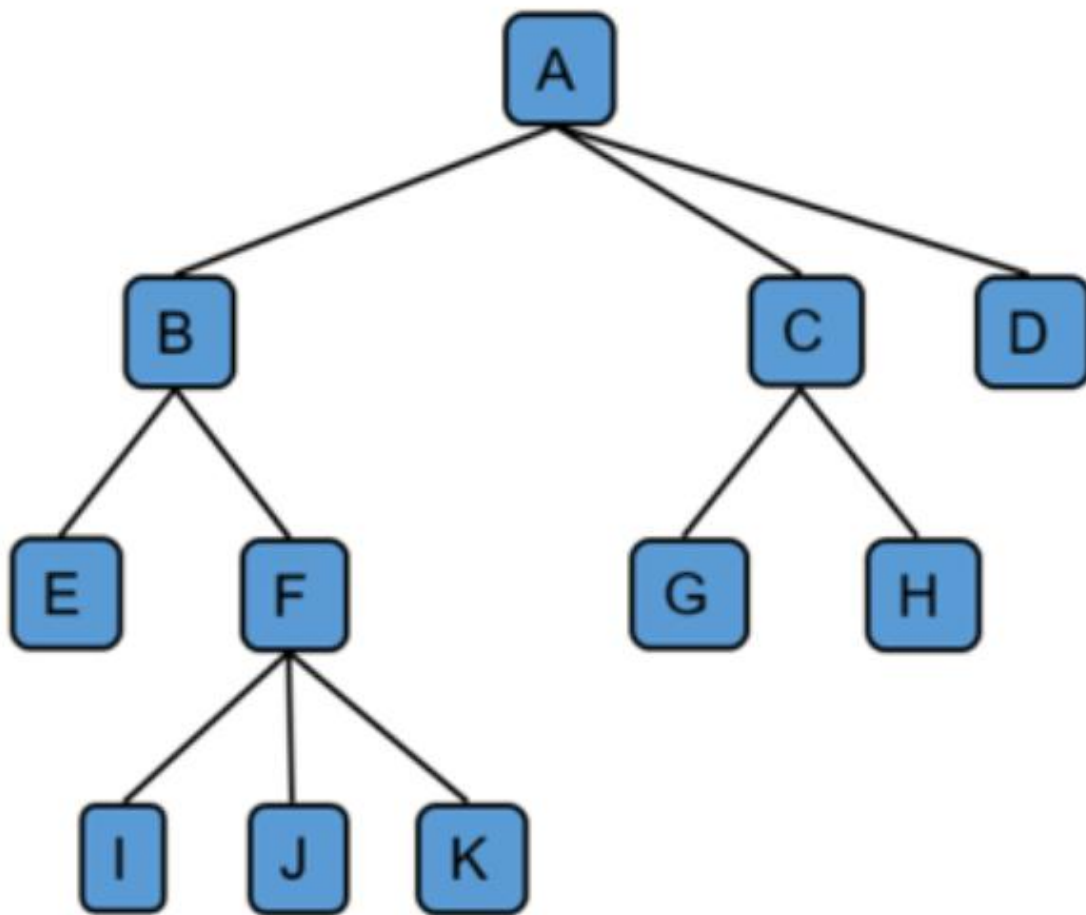


Inorder: 4, 6, 3, 7, 2, 5, 1

Preorder: 7, 6, 4, 3, 5, 2, 1

Postorder: 4, 3, 6, 2, 1, 5, 7

Question No 10



Inorder: E, B, I, F, J, A, G, C, H

Preorder: A, B, E, F, I, J, C, G, H

Postorder: E, I, J, F, B, G, H, C, A